Zhaoqi Li

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EDUCATION

Ph.D. Statistics, University of Washington 06/2024
Thesis: Estimation and Inference of Optimal Policies

B.A. Mathematics and Computer Science, Macalester College, Saint Paul, MN 05/2018
Summa Cum Laude, GPA 3.98/4.00

RESEARCH INTEREST

Reinforcement learning, causal inference, bandits, policy learning, adaptive experimental design

RESEARCH AND WORK EXPERIENCE

STANFORD UNIVERSITY, Stanford, CA 09/2024 – Current
Postdoctoral Scholar, Advisor: Emma Brunskill

Project 1: Efficient Tests for the Benefit of Personalizing Interventions

- Design an efficient hypothesis test to assess whether personalization yields utility in a given domain
- Establish semiparametric efficiency (minimum possible asymptotic variance among all estimators) of the estimator for utility of personalization formally
- Show that our approach matches or exceeds prior approaches in accuracy and Type-I error in real datasets on healthcare, job training, education, and joke recommendations
- Manuscript submitted to *Science*

Project 2: Active Learning for Stochastic Contextual Linear Bandits

- Develop an algorithm for SCLB that actively sample contexts and actions to learn a near-optimal policy
- Prove that our algorithm achieves the same worst-case simple regret as passive learning
- Demonstrate substantial gains over passive learning in certain instances both theoretically and in experiments on warfarin dosage prescription and joke recommendation
- Paper submitted to *AISTATS*

UNIVERSITY OF WASHINGTON, Seattle, WA 09/2018 – 06/2024
Doctoral Student Researcher, Advisors: Lalit Jain, Alex Luedtke

Project 1: Instance-optimal PAC Algorithms for Contextual Bandits

- Established the first instance-dependent lower bound for the sample complexity of probably approximately correct (PAC) policy identification in contextual bandit
- Designed a computational efficient algorithm and formally proved its instance optimality by matching the sample complexity lower bound

Project 2: Optimal and Efficient Algorithm for Best-arm Identification in Linear Bandits

- Developed an instance-optimal algorithm for best-arm identification in linear bandits and formally proved its instance optimality
- Matched the same level of simplicity as Thompson sampling by showing that our algorithm only needs sampling oracles
- Showed that our algorithm matches or outperforms existing benchmarks in terms of sample complexity in various simulation instances

Project 3: Efficient estimators for subsidiary metrics under an optimal policy

- Developed an estimator for the performance of the optimal policy on other metrics to address the challenge of multiple metrics of consideration in policy learning
- Established semiparametric efficiency (minimum possible variance) of our estimator under a margin condition
- Proposed two complementary approaches (two-stage and joint) that provide confidence intervals for the subsidiary metrics performance in general settings
- Demonstrated via numerical simulations that the confidence interval of the joint approach is at most twice as wide as the tightest possible confidence interval

META INC., Seattle, WA 06/2024 – 09/2024
Research Scientist Intern

Project: Effect Size Estimation

- Developed a novel, non-parametric SIMEX-based estimator to recover the distribution of true effect sizes from noisy A/B-test estimates
- Integrated isotonic regression to enforce monotonicity of the estimated CDF
- Proved our estimator is consistent when the true effect distribution is parametric
- Validated the approach via simulations by showing it matches or outperforms the empirical Bayes normal-means benchmark across various effect-size scenarios

AMAZON INC., Seattle, WA 06/2021 – 09/2021
Applied Scientist Intern

Project: Adaptive Experimental Design for Time Variation

- Analyzed production data to demonstrate the existence of time variation phenomenon
- Designed bandit algorithms with change-point detection to help with automated decision-making under time variation
- Demonstrated using the Amazon data that our approach outperforms existing Thompson sampling benchmark in terms of cumulative regret and effectively detects time variation

AMAZON INC., Seattle, WA 06/2020 – 09/2020
Applied Scientist Intern

Project: Structured Multivariate Testing

- Implemented belief propagation algorithm for graph optimization to improve the multivariate testing framework in production
- Demonstrated that belief propagation matches or outperforms the current hill climbing and exhaustive search baselines via simulation analysis with different cases

PUBLICATIONS

Journal Publications

Z. Li, H. Nassif, A. Luedtke (2025). Estimation of subsidiary performance metrics under an optimal policy. *Statistica Sinica*.

L. Jain, **Z. Li**, E. Loghmani, B. Mason, H. Yoganarasimhan (2024). Effective Adaptive Exploration of Prices and Promotions in Choice-Based Demand Models. *Marketing Science*.

J. A. Rathe, E. A. Hemann, J. Eggenberger, **Z. Li,** M. L. Knoll, C. Stokes, T. Y. Hsiang, J. Netland, K. K. Takehara, M. Pepper, M. Gale Jr (2021). SARS-CoV-2 Serologic Assays in Control and Unknown Populations Demonstrate the Necessity of Virus Neutralization Testing. *The Journal of infectious diseases*, 223(7), 1120–1131.

B. Anzis, S. Chen, Y. Gao, J. Kim, **Z. Li,** R. Patrias (2018). Jacobi-Trudi determinants over finite fields. *Annals of Combinatorics*, 22(3), 447-489.

Y. Gao, **Z. Li**, T. Vuong, L. Yang (2018). Toric Mutations in the dP2 Quiver and Subgraphs of the dP2 Brane Tiling. *The Electronic Journal of Combinatorics,* 26(2), P2-19.

Conference Publications

Z. Li, K. Jamieson, L. Jain (2024). Optimal Exploration is no harder than Thompson Sampling. *International Conference on Artificial Intelligence and Statistics.* PMLR, 2024.

Z. Li, L. Ratliff, H. Nassif, K. Jamieson, L. Jain (2022). Instance-optimal PAC algorithms for contextual bandits. *Advances in Neural Information Processing Systems,* 35, 37590-37603.

Submitted and Preprints

Z. Li, E. Brunskill (2025). A Statistical Test for the Benefits of Personalizing Interventions. *Submitted to Science.*

I. Karmarkar, **Z. Li**, E. Brunskill (2025). Active Learning for Stochastic Contextual Linear Bandits. *Submitted to AISTATS.*

E. Bakhitov, A. Singh, **Z. Li** (2025). Causal Gradient Boosting: Boosted Instrumental Variable Regression. *Under major revision for JMLR.*

Z. Li, Y. Ma, C. Vajiac, Y. Zhang (2018). Exploration of Numerical Precision in Deep Neural Networks. *arXiv preprint arXiv:1805.01078.*

In preparation

Z. Li, D. Ting, E. Emamjomeh-Zadeh, H. Nassif. Estimating the true effect size distribution with simex.

PRESENTATIONS

Efficient tests for the Benefit of Personalizing Interventions, Stanford Causal Science Center Conference on Experimentation, May 2025

Efficient Statistical Tests for Personalization Effects, Stanford Data-driven Seminar, May 2025

Efficient Statistical Tests for Personalization Effects, Stanford ML Seminar, May 2025

Optimal Exploration is no harder than Thompson Sampling, INFORMS invited talk, October 2024

Estimation and Inference of Optimal Policies, PhD final exam, May 2024

Optimal Exploration is no harder than Thompson Sampling, AISTATS, May 2024

Optimal Exploration is no harder than Thompson Sampling, INFORMS invited talk, October 2023

Instance-optimal PAC Algorithms for Contextual Bandits, NeurIPS, December 2022

Instance-optimal PAC Algorithms for Contextual Bandits, INFORMS invited talk, October 2022

Instance-optimal PAC Algorithms for Contextual Bandits, Amazon Prime Science Seminar, November 2021

LEADERSHIP EXPERIENCE AND SERVICE

REVIEWER

Conference on Neural Information Processing Systems (NeurIPS) 2022 – 2025

International Conference on Artificial Intelligence and Statistics (AISTATS) 2022 – 2024

Conference on Uncertainty in Artificial Intelligence (UAI) 2024

Journal of the American Statistical Association (JASA) 2024

DEPARTMENTAL SERVICE

Stanford AI4HI Reading Group Organizer 2024 – Current

Reviewing PhD applications for the Department of Statistics at the University of Washington

2020, 2021, 2023

TEACHING AND MENTORING EXPERIENCE

DEPARTMENT OF COMPUTER SCIENCE, STANFORD UNIVERSITY

Student Mentor 03/2025 - Present

- Mentor an undergraduate student in a research project on developing a simple and optimal algorithm for PAC-policy learning in contextual bandits aimed at practical use
- Guide the student in understanding the research topic, brainstorming ideas, and deriving theoretical details through weekly meetings and email communications

DEPARTMENT OF STATISTICS, UNIVERSITY OF WASHINGTON

Student Mentor 01/2020 – 12/2020

- Mentored two undergraduate students on two projects on Statistical Machine Learning and Data Analysis and Statistical Illusions in the Directed Reading Program
- Designed the entire curriculum for the Statistical Illusions project
- Prepared reading materials and led meetings each week
- Advised the final presentation and write-ups

DEPARTMENT OF STATISTICS, UNIVERSITY OF WASHINGTON

Teaching Assistant 09/2018 – 06/2021

STAT 391: Quantitative Introductory Statistics for Data Science 2021

STAT 341: Introduction to Probability and Mathematical Statistics II 2021

STAT 535: Statistical Learning: Modeling, Prediction, and Computing 2019 - 2021

STAT 529: Sampling Survey Techniques 2019

STAT 221: Statistical Concepts and Methods for the Social Sciences 2019

STAT 340: Introduction to Probability and Mathematical Statistics I 2018

- Taught discussion sections and prepared the materials for these sections
- Wrote homework and exam questions with solutions
- Held office hours, graded homework and exams, and assisted the instructor with class

DEPARTMENT OF STATISTICS, UNIVERSITY OF WASHINGTON

Lead Tutor 09/2020 – 09/2021

- Acted as point of contact for tutoring requests for the Statistics Tutor and Study Center
- Hired and trained six new tutors including the design of training curriculum and hosting practice sessions
- Organized the quarterly schedule of tutors and assisted the director of tutoring center

DEPARTMENT OF STATISTICS, UNIVERSITY OF WASHINGTON

Tutor 09/2019 – 06/2020

- Hosted drop-in tutoring sessions in the Statistics Tutor and Study Center
- Supported other tutors answering questions from advanced undergraduate and graduate classes